

Portfolio

Michelle Liu

2025 Selected Works

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Spring 2023

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Annum Architects
2023 - 2024

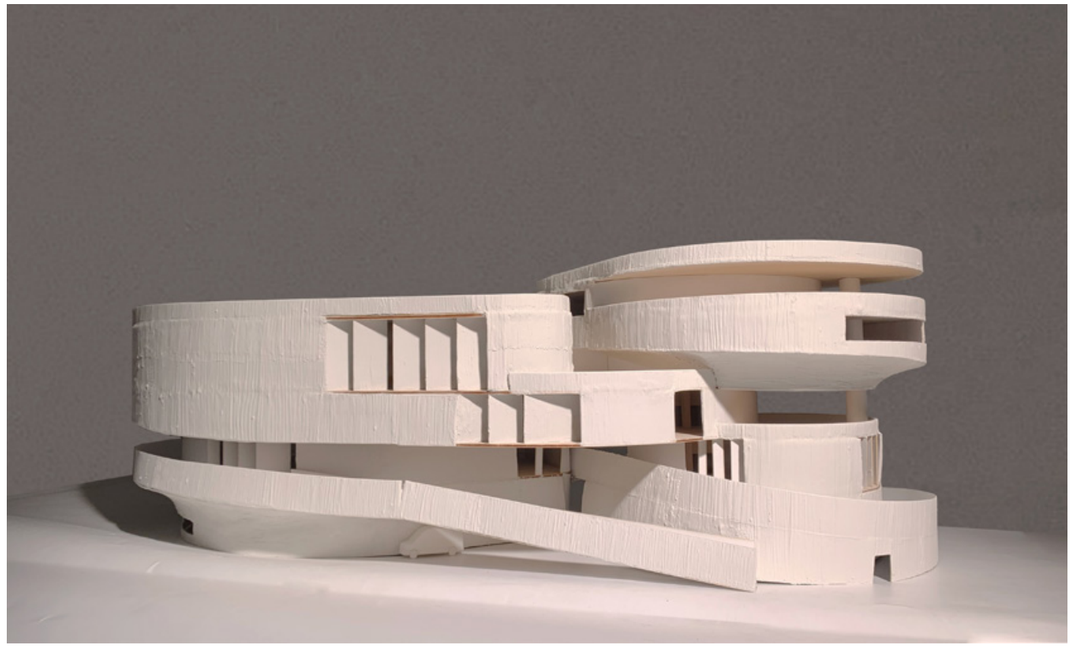
01. **Soft Entwine**

Course: ARC2014 - Design Studio IV
Instructor: Chris Cornecelli

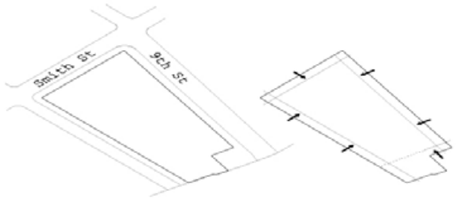
Continuity invites curiosity. Bridging together art and community through continuous circulation as program, users are brought together to experience the various production stages of creation. Through the weaving of spaces, Soft Entwine proposes an architecture of circulation that is both playful and multifaceted.

Located in Brooklyn, New York at 503 Smith Street, the site along the Gowanus canal embodies the magnitude and urgency of the climate threat in an area transitioning away from a heavy industrial past into an eco-industrial and mixed use future. The project aims to reimagine the social and material dimensions of buildings to define the urgent transition of the industry into a low carbon and circular economy. It is a social infrastructure for the neighborhood anchored around a library of things, a reuse factory and also an exhibition and workspace space for the local artists' community.

Collaborator: Amelia Chung

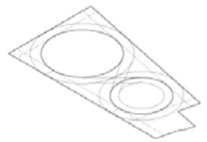


1:100 Model



existing site

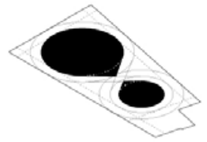
offset



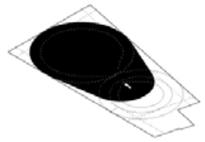
connect



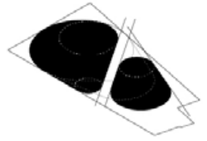
ground ramp



F1+2 massing



F3 massing

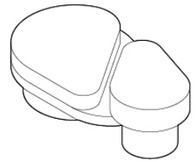


F4 massing



above ground ramp

Design Development



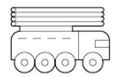
1. Design



2. Local Sourcing

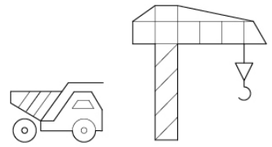


Raw Materials





Manufacturer

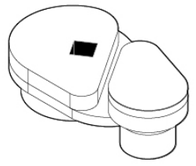


3. Construction

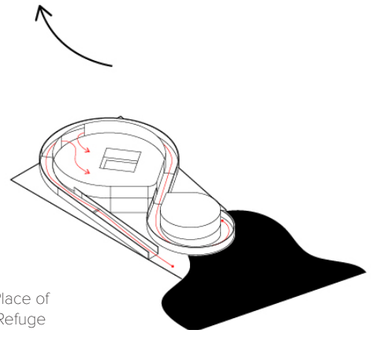


Use of low impact building materials

Local CLT Manufacturer



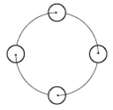
4. Operation and Public Use



Place of Refuge

Flood Mitigation

Mix Program Use



5. Renew

Flexibility + Adaptability



Modularity

Alternative Assemblies

6. Disassembly

7. Repurpose

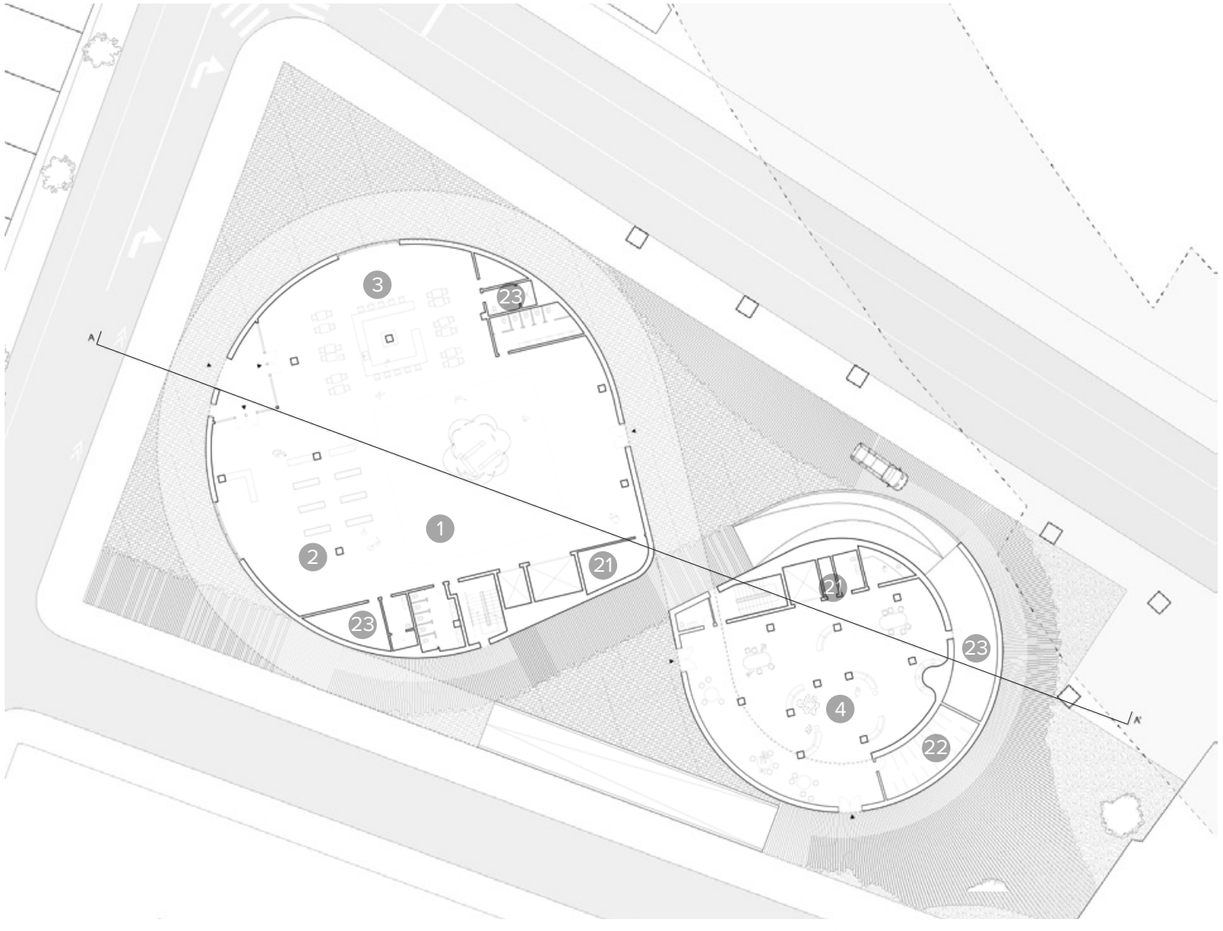


Material Recycling

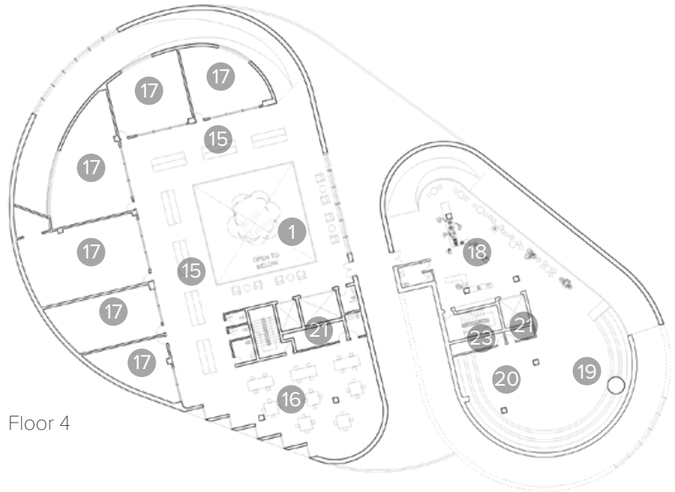
Material Reuse

Life Cycle Manifesto

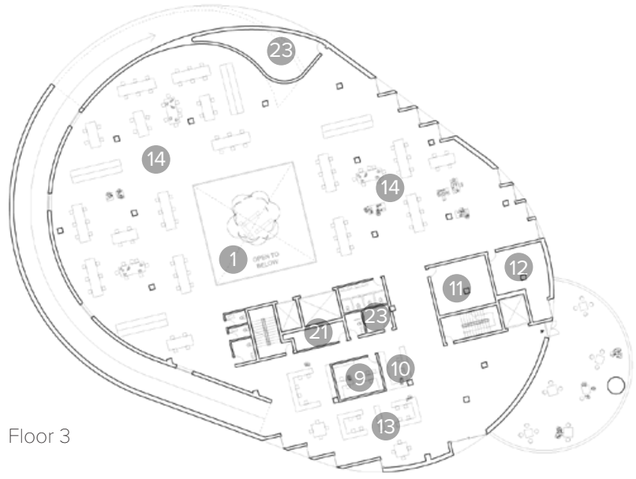
1 Art Exhibition
2 Cafe / Restaurant
3 Circular Store
4 Play Area
5 Loading Area
6 Material Storage
7 Artist Studio
8 Start-Up Office
9 Librarian Office
10 Reference Desk
11 Staff Refurbishing Shop
12 Staff Area
13 Sharing Depot
14 Workshop Area
15 Artist Space Lockers
16 Repair Cafe
17 Training, Making, Repair Workshops
18 Community Kitchen / Catering
19 Auditorium / Multi-Purpose Space
20 Main Stage
21 MEP
22 Bike Storage
23 Storage



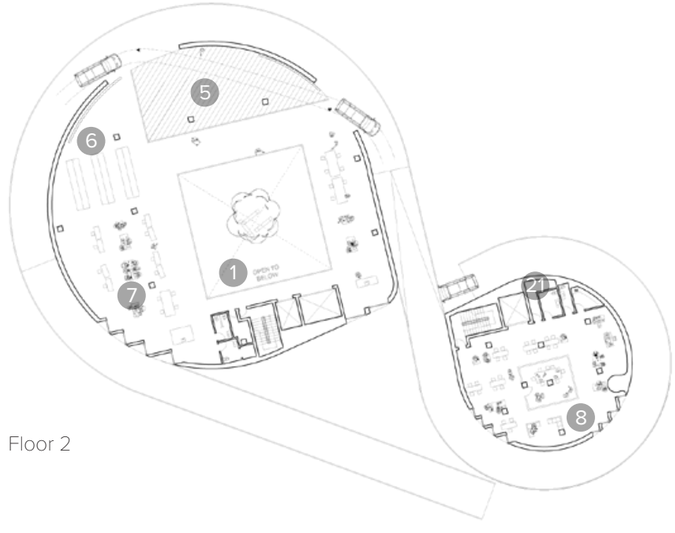
Ground Floor Plan



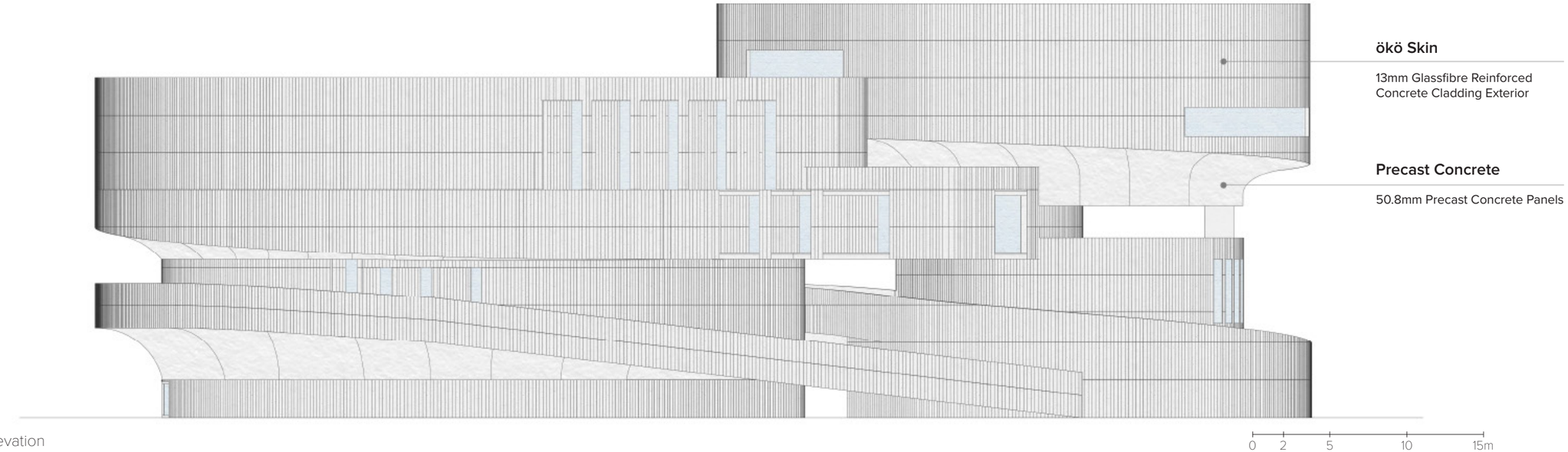
Floor 4



Floor 3



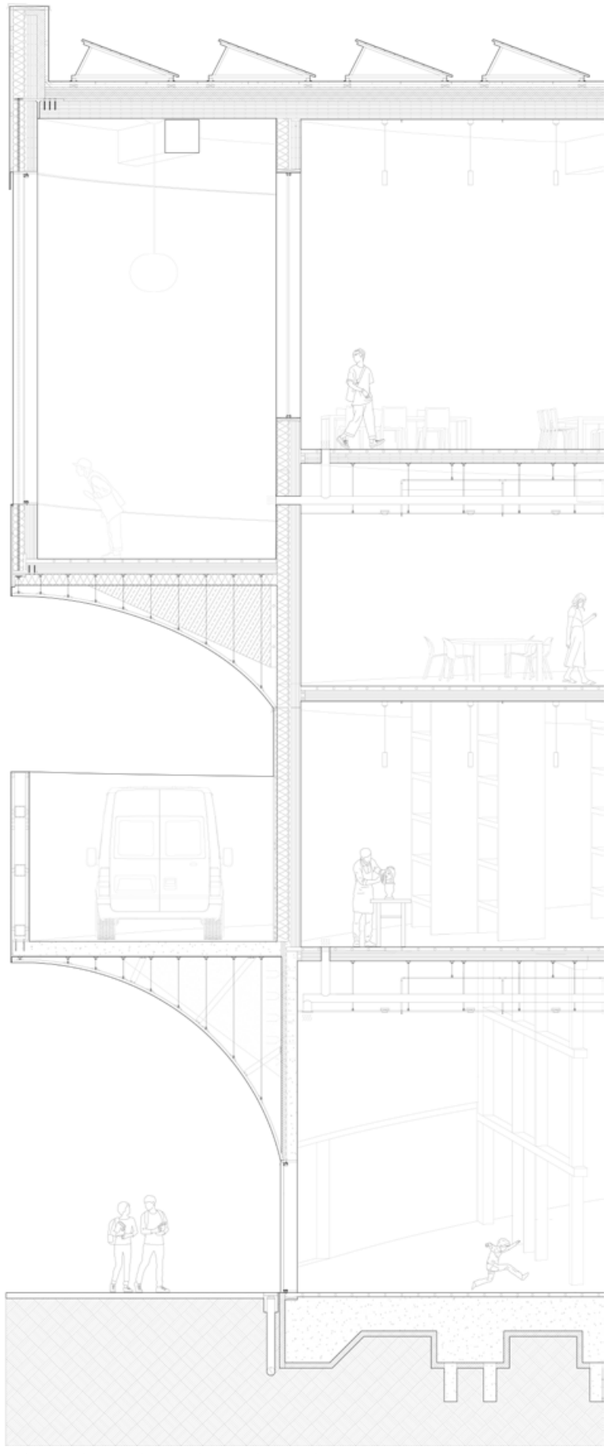
Floor 2



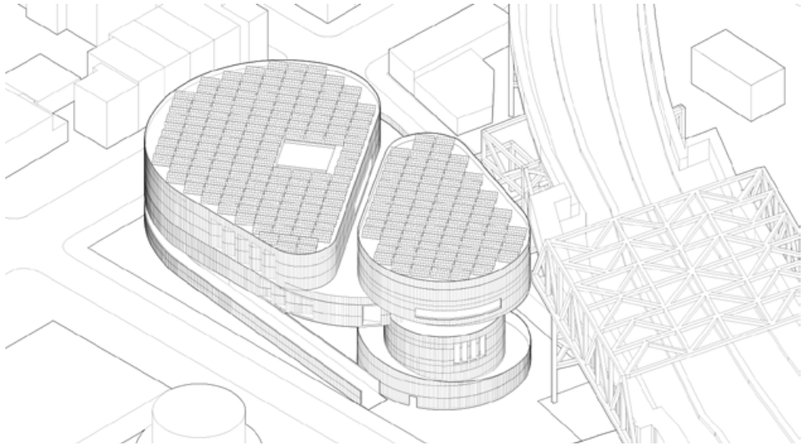
South Elevation



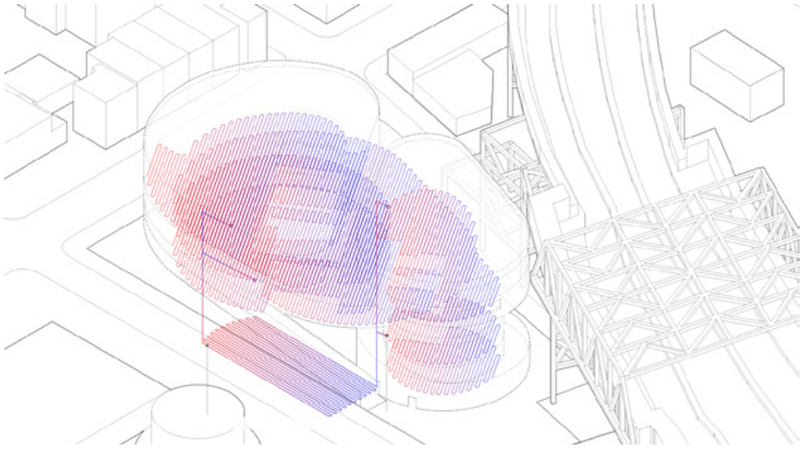
Section A-A'



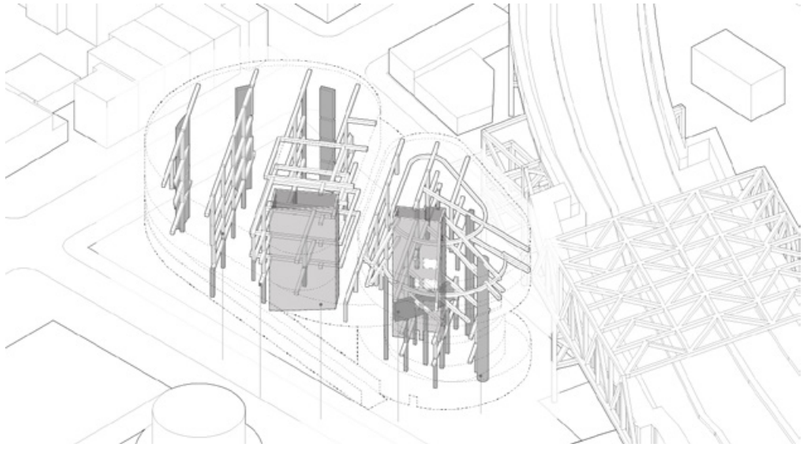
Detail Section



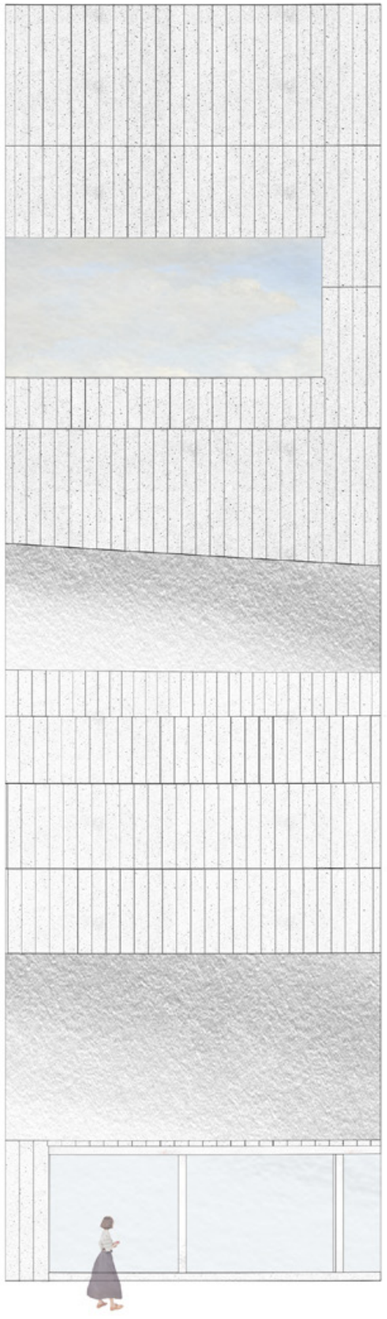
Renewable Energy Sources - Roof Mounted PV Panels



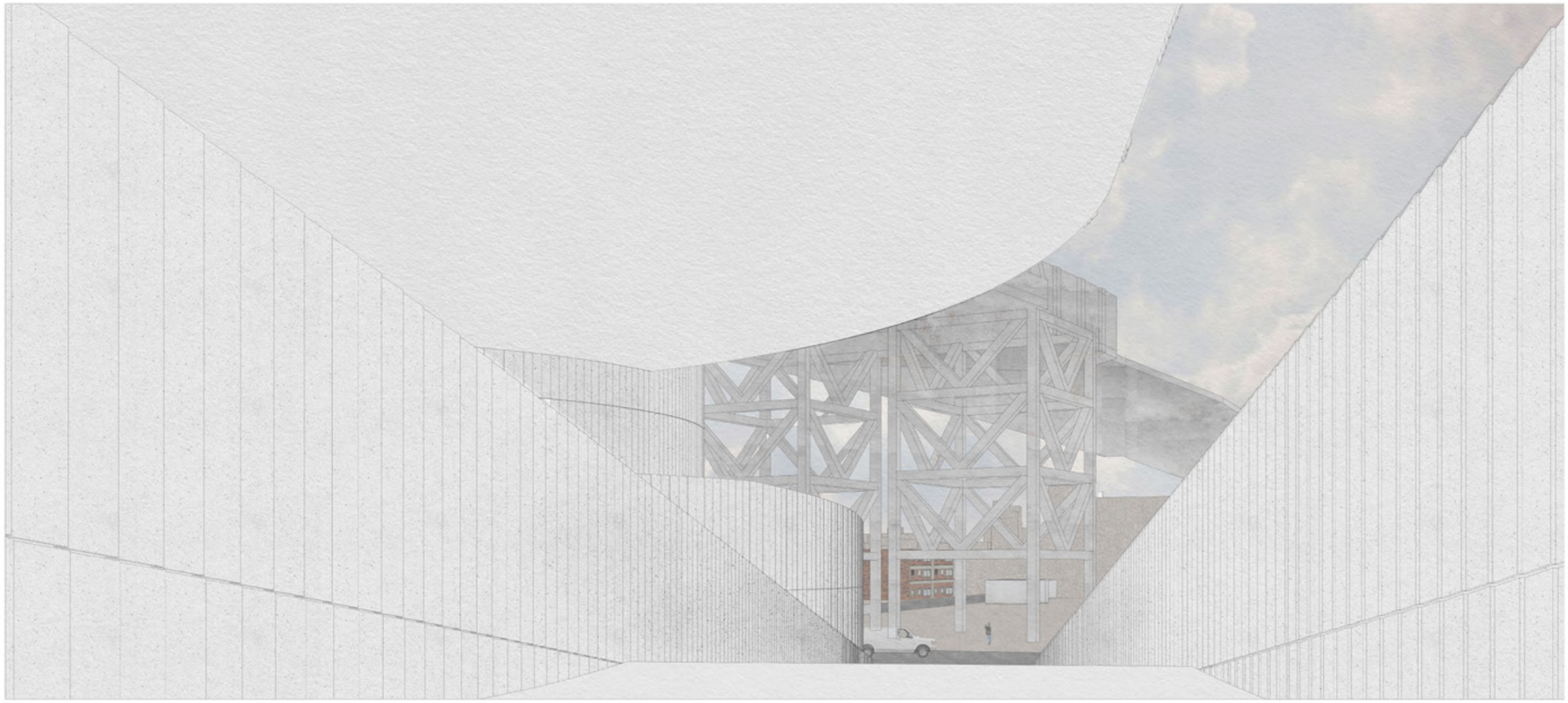
Renewable Energy Sources - Hydronic Radiant Heating and Cooling



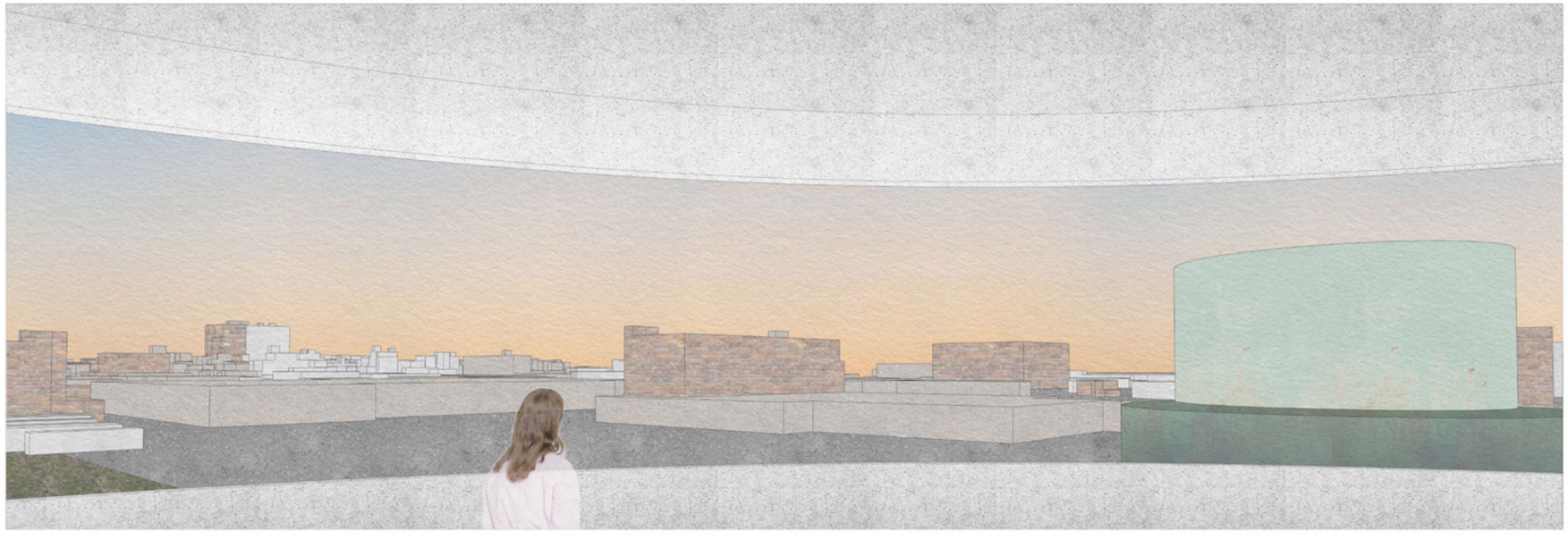
Structural System - Mass Timber & Concrete Core



North Entrance - Collage



Auto Ramp Entrance - Collage



South Balcony - Collage

02. **Big Little Village**

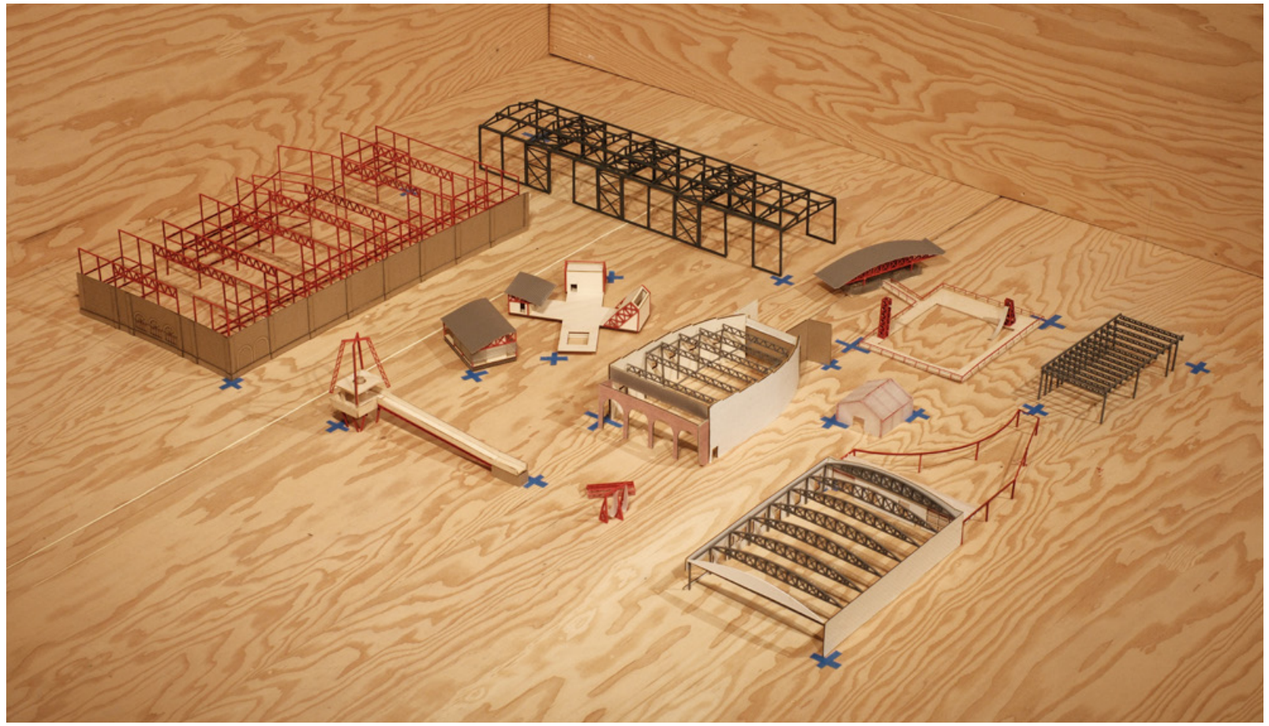
Course: ARC3015 - Option Studio
Instructor: Florian Idenburg and Jing Liu of SO-IL

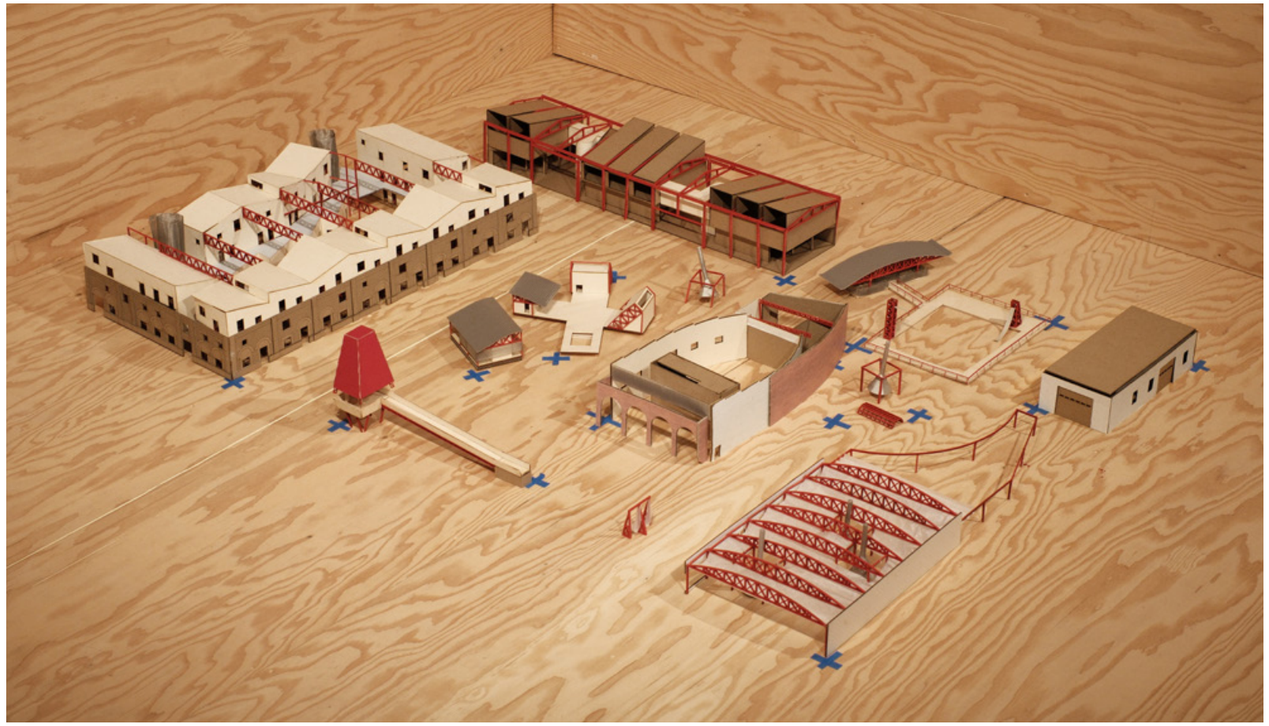
Big Little Village reimagines a strip of land in Detroit, between Jefferson Avenue and the Detroit River, as a site of care, repair, and transformation. Rooted in the city's ethos of resilience and resourcefulness, the project catalogues all existing materials on site—buildings, surfaces, and found objects—as a foundation for design. By celebrating what's already there, it proposes adaptive reuse strategies that prioritize repair over demolition and evolution over replacement, repairing memory through architecture and honoring the histories embedded in place.

Framed around three "states of being"—Tending, Invitation, and Becoming—the project resists linear development in favor of temporal shifts that respond to changing community needs. Each building operates as an autonomous yet interconnected entity, supporting acts of play, creation, rest, and gathering in structures that once served as a theatre, boat storage, and navy yard.

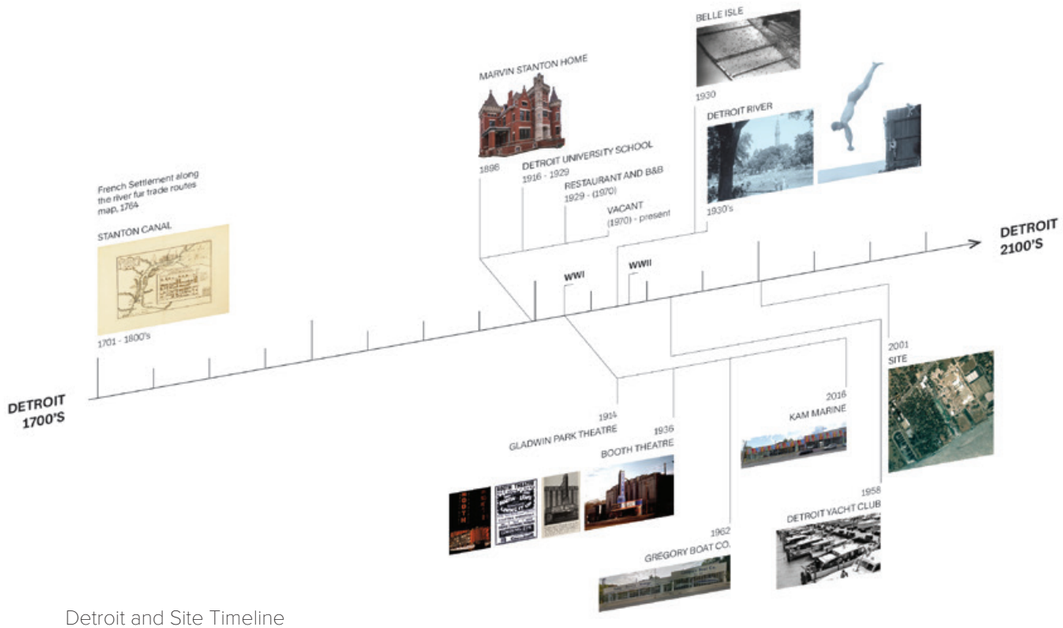
A key focus is the site's reconnection to the larger community, inviting public interaction with water through improved riverfront access. This gesture addresses historical inequities while reaffirming Detroit's deep ties to its waterfront. Balancing preservation with progress, the project becomes a slow, collective act of making—a living archive that grows with its people. It is a vision rooted in resilience, committed to care, and unapologetically Detroit.

Collaborators: Nichola Basford, Ariana Fernandez Chesquin, Declan Roberts





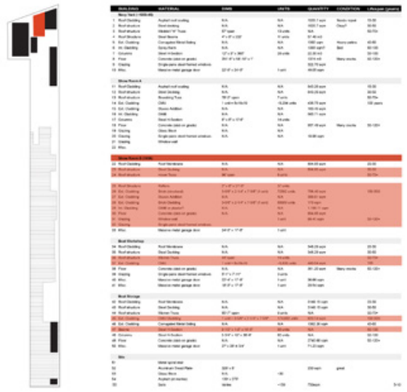
1:100 Models Captured at Various States of Being



Detroit and Site Timeline



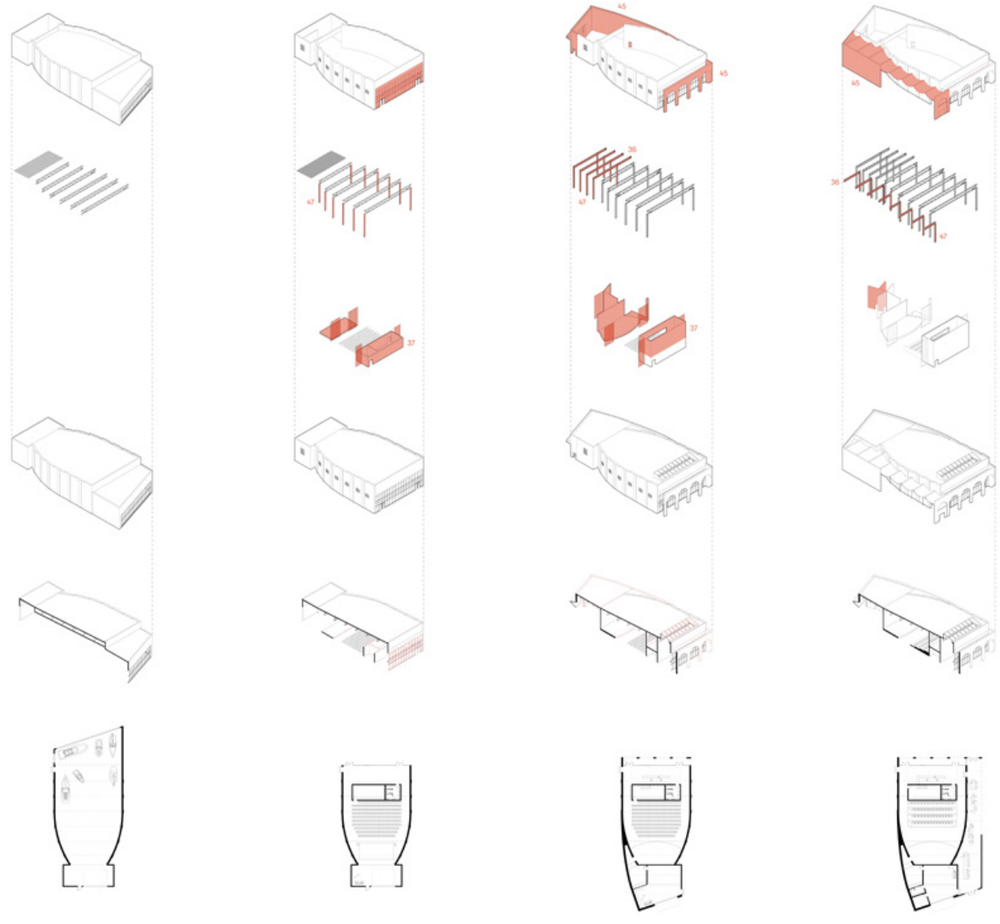
Existing Material on Site



Existing Material Catalogue - Theatre



Theatre Collage in Various States of Being

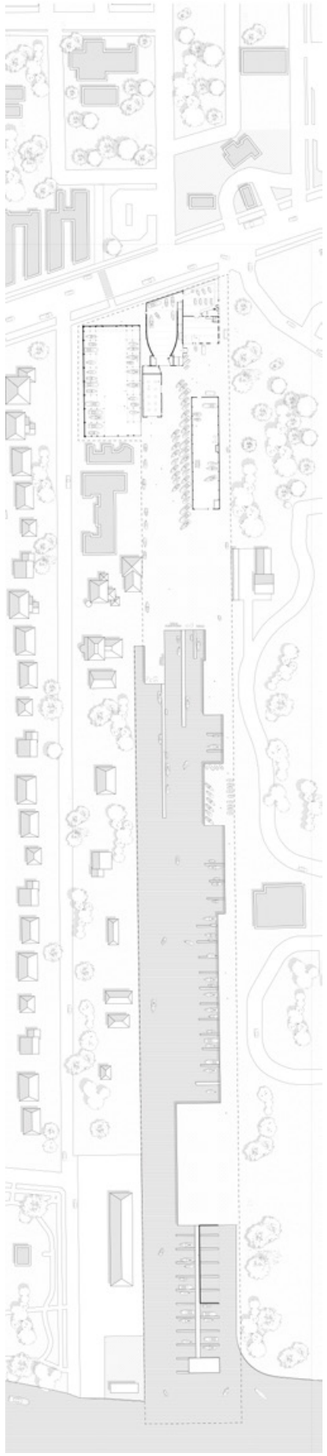


Existing Theatre Builing
(Boat Storage)

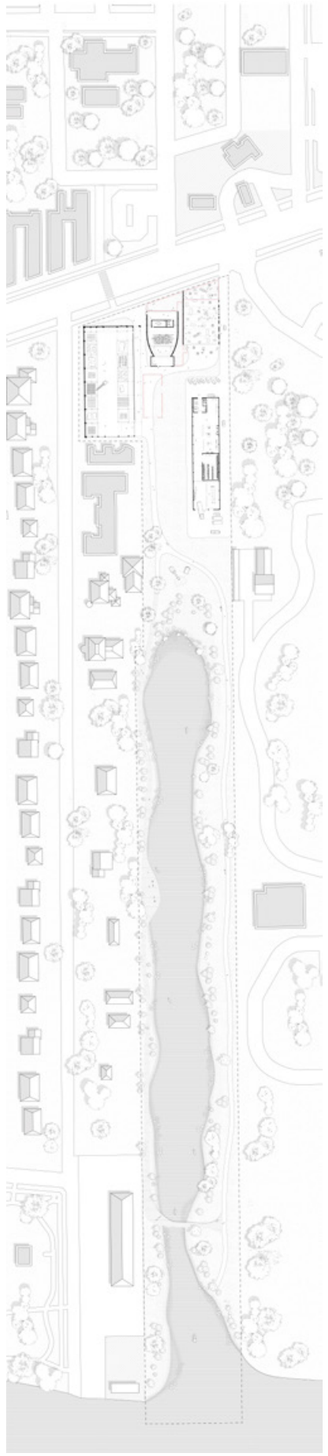
Tending
(Community Arts Center)

Inviting
(Formal Theatre)

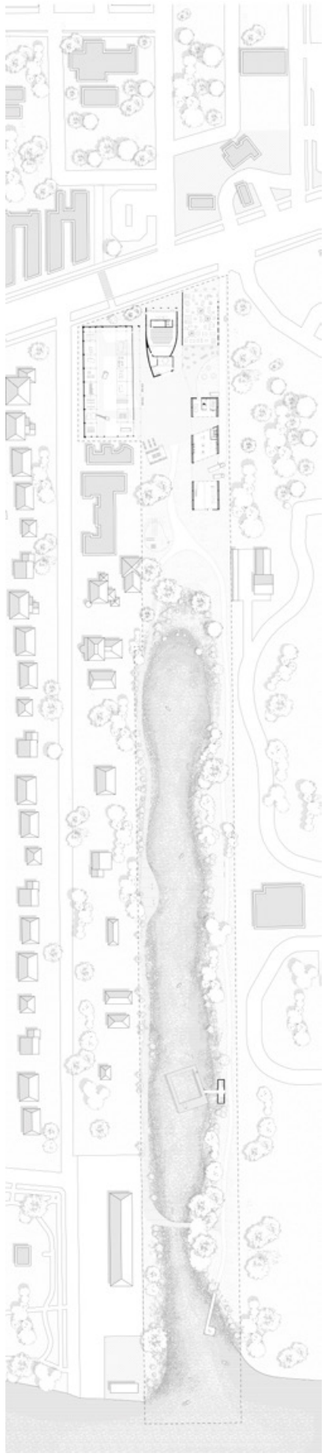
Becoming
(Movie Theatre + Bar)



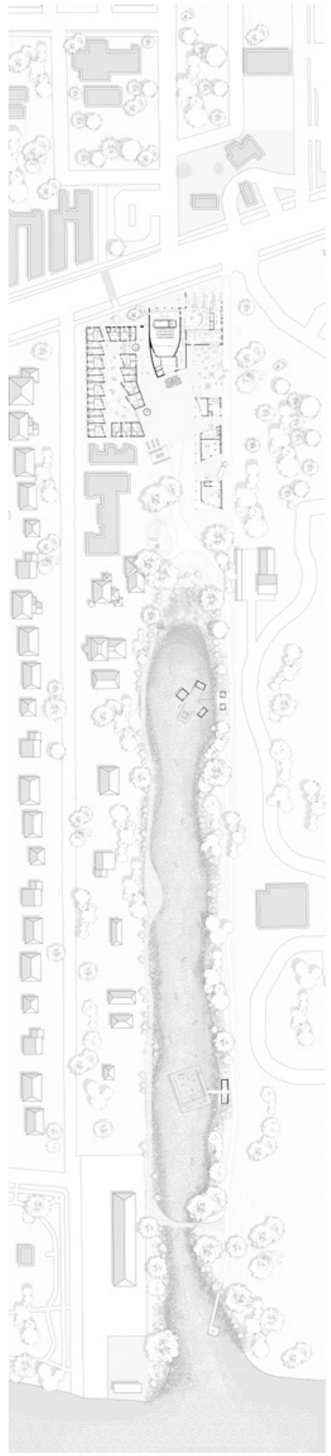
Site Plan - Existing



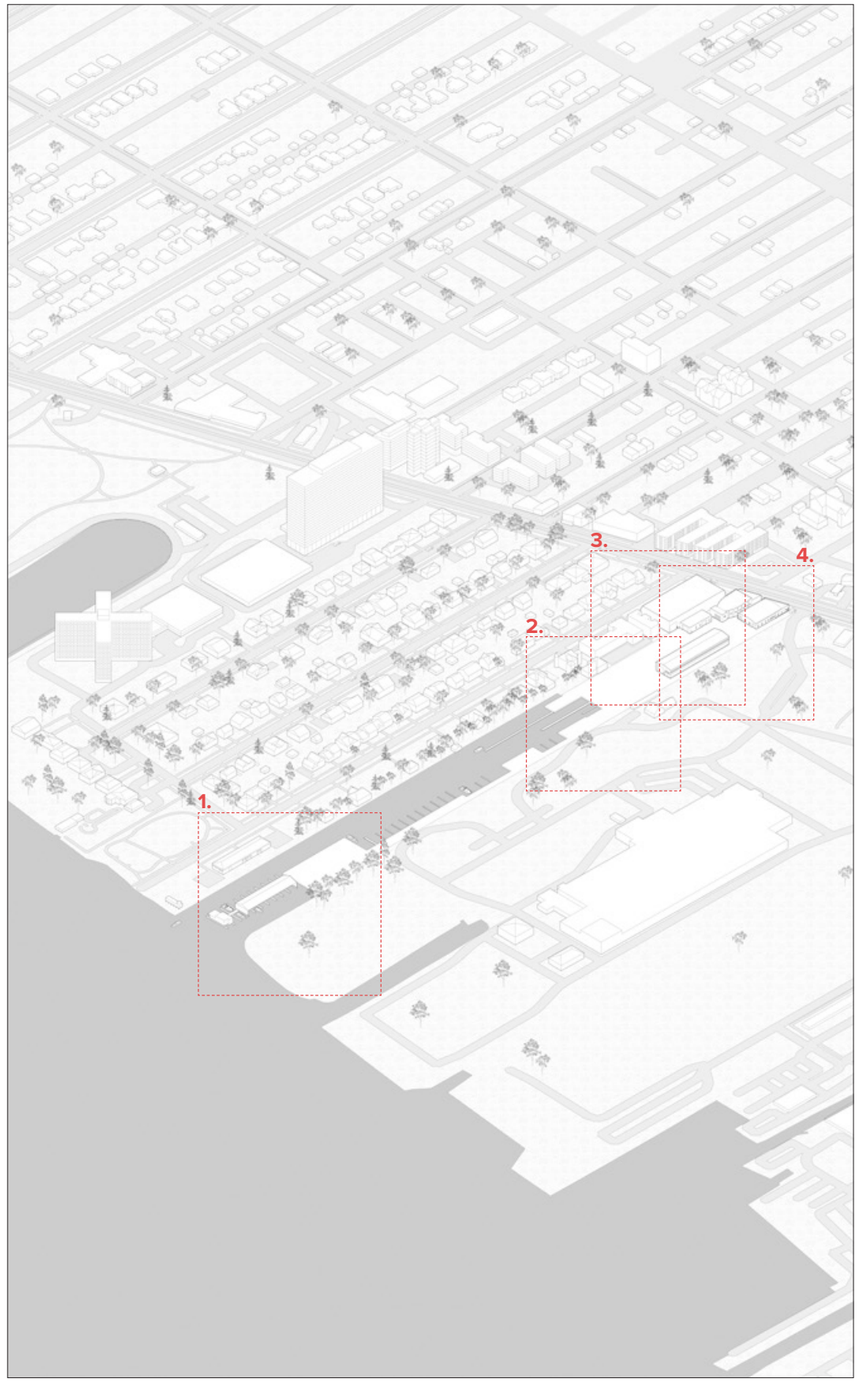
Tending



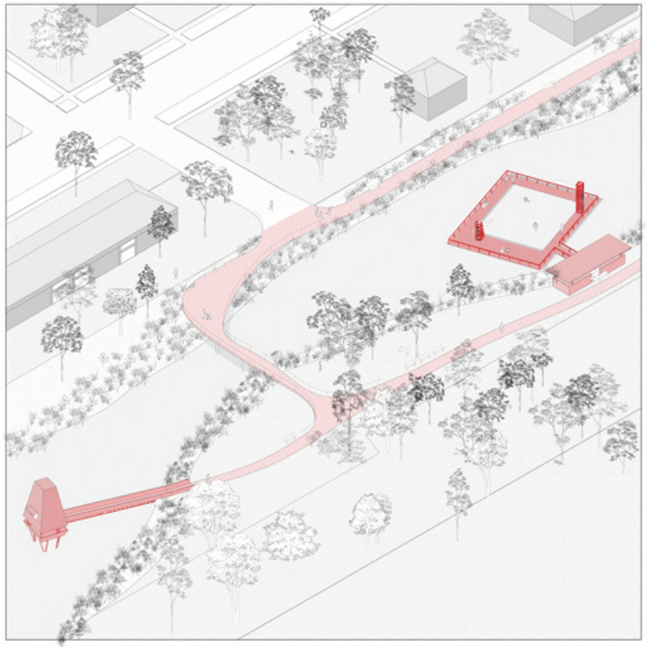
Inviting



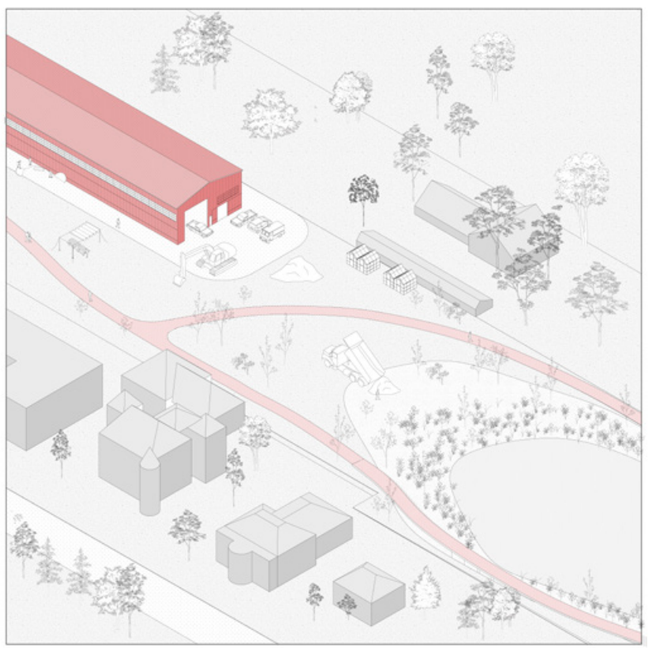
Becoming



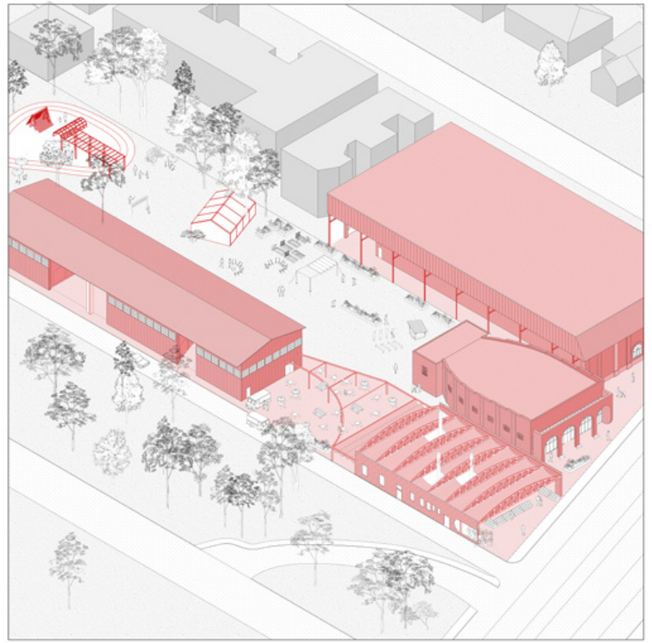
Axo



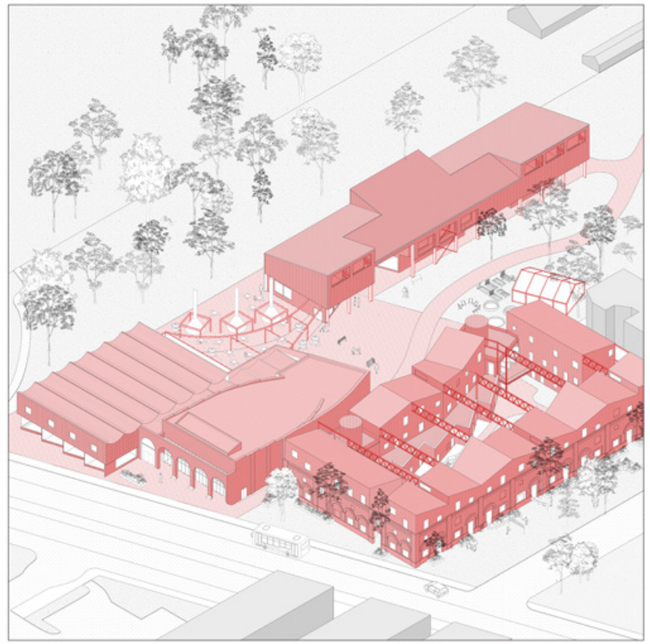
1. Inviting - Bird House + Community Pool Addition



2. Tending - Site Remediation + Material Processing



3. Inviting - Market Hall and Workshop Hub + Playground Addition



4. Becoming - Formal Housing, Theatre, and Restaurants

03. Church of Community

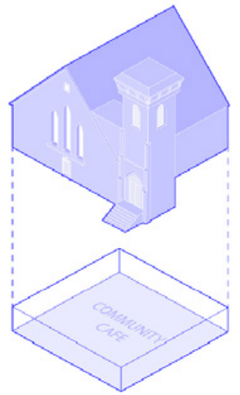
Course: ARC1041 - Building Science I
Instructor: Ted Kesik

Historically, churches have served as communal gathering spaces, but with declining attendance, many have become underutilized or abandoned. This project repurposes Headford United Church in Richmond Hill, Ontario, transforming it into a community hub that fosters inclusivity and shared resources.

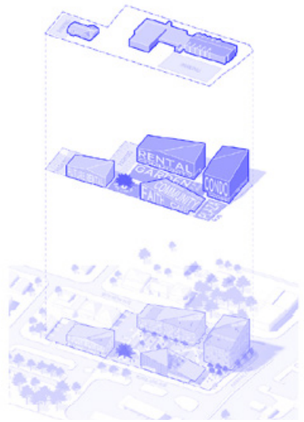
By introducing an urban garden, sanitation facility, tool-sharing workshop, and communal kitchen, the design provides essential services that bridge generational and socio-economic gaps. These programs create accessible spaces for learning, working, and gathering, reinforcing a support network within the community. As a replicable model for adaptive reuse, the project demonstrates how abandoned buildings can be revitalized into active, community-driven spaces.

These programs are necessary to create affordable community-based initiatives for the public to have access to shared spaces, tools, and food. The proposed new typologies within this project can be replicated and adapted to other underutilized or abandoned buildings and serve as a precedent for future projects of similar size.

Collaborators: Jessica Babe, Amelia Chung, Larissa Ho, Rachel Sau



1.0 St. John's Anglican Church
Chapleau, ON



2.0 St. Julian of Norwich Church
Ottawa, ON

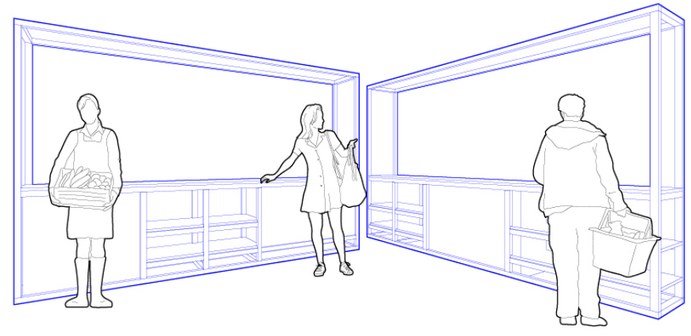


3.0 Cecil Community Centre
Toronto, ON

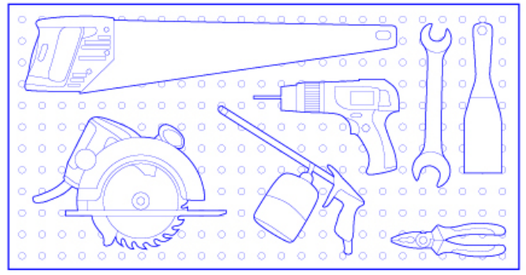
Precedent Research



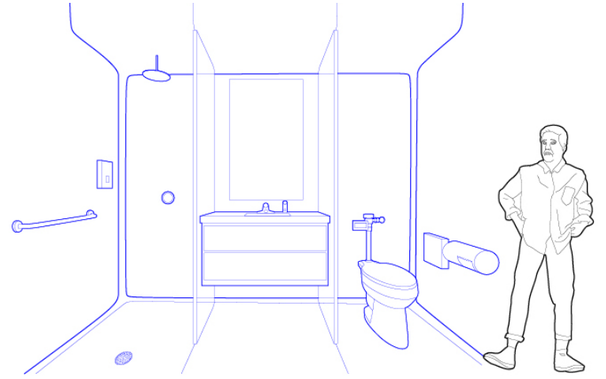
1.0 Urban Garden — Accessible raised garden beds



3.0 Community Kitchen Facility — Design by Himani Harikrishna Ravuri

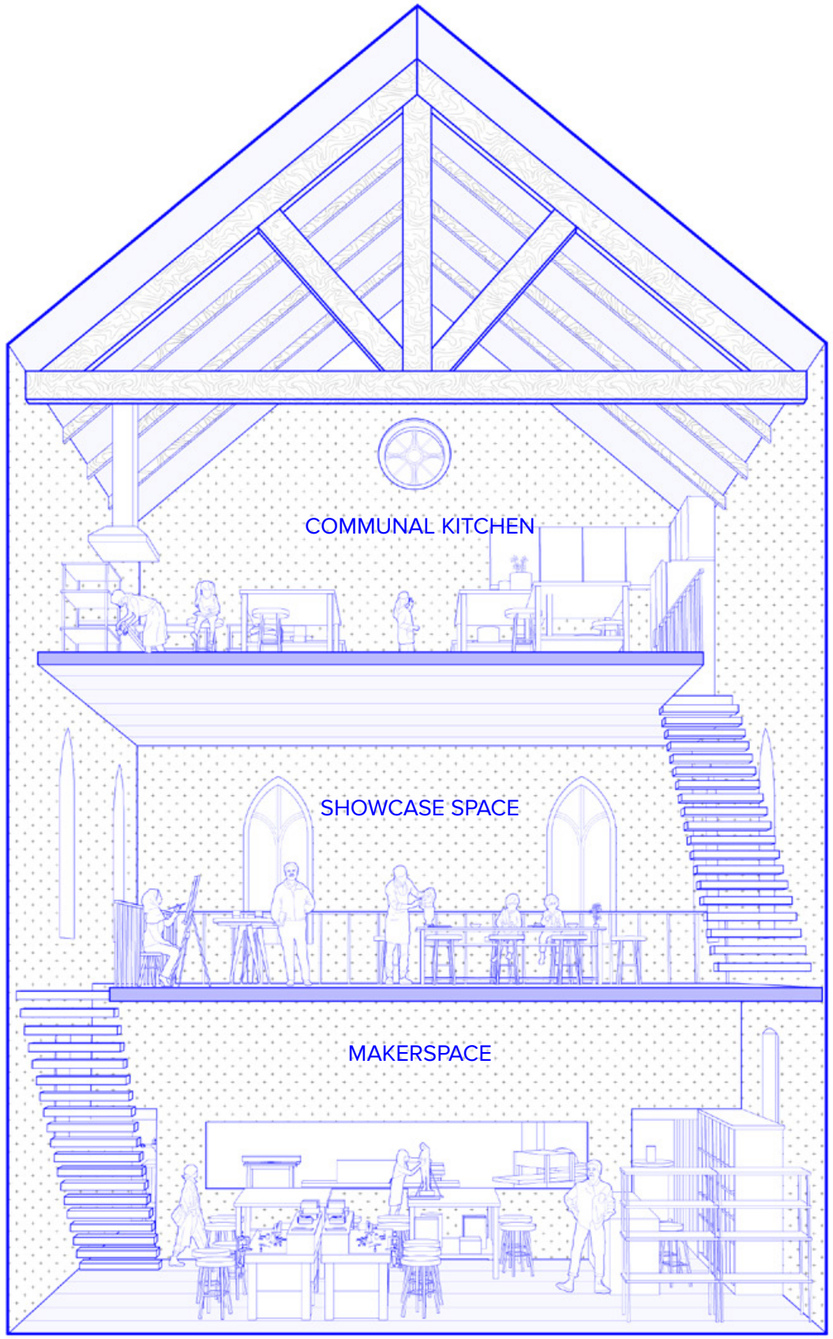


2.0 Tool Sharing & Workshop — Toronto Tool Library

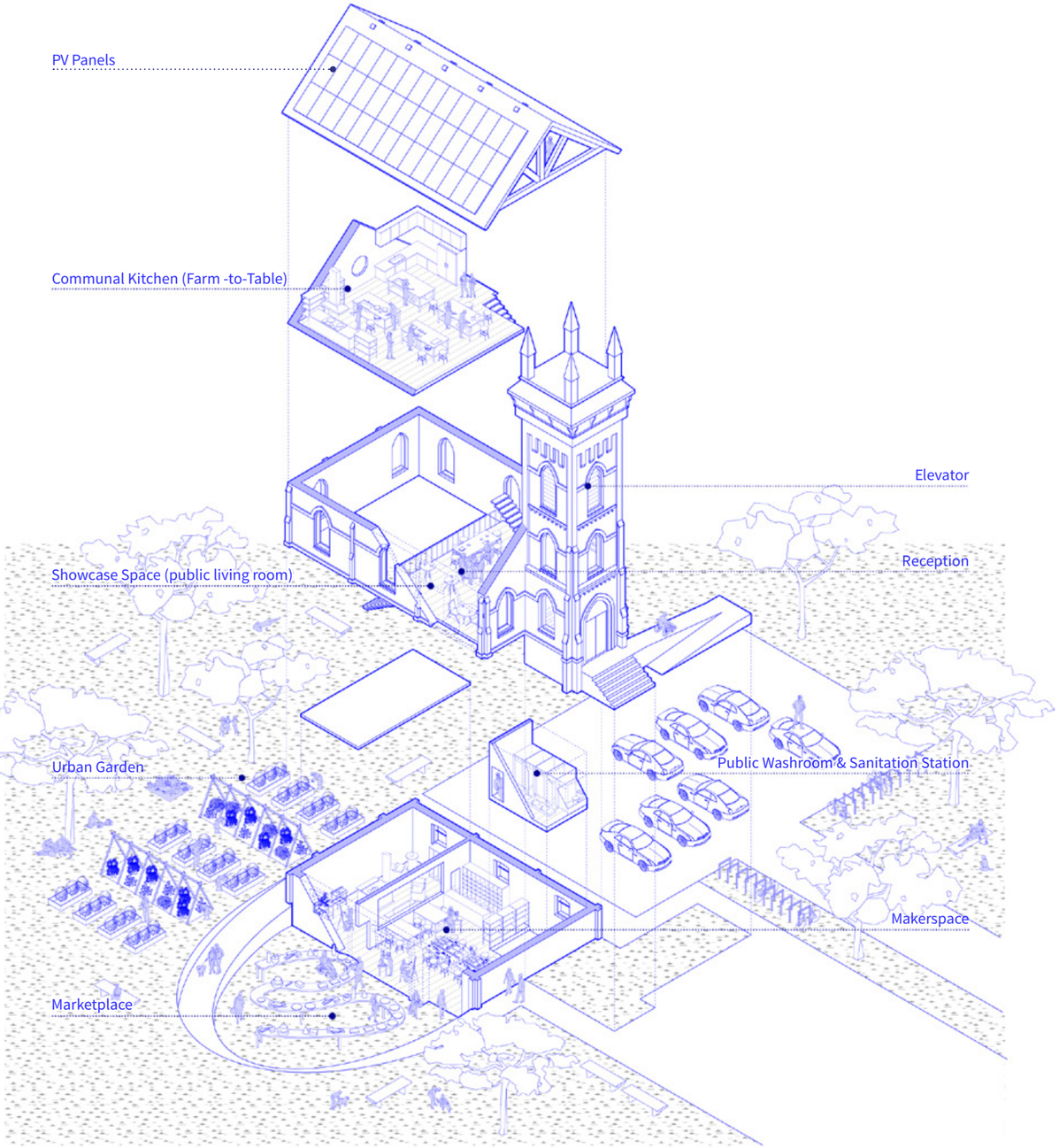


4.0 Sanitation Station — StudioAC mobile shower facility design

Proposed Programs



Interior Perspective



Exploded Program Diagram

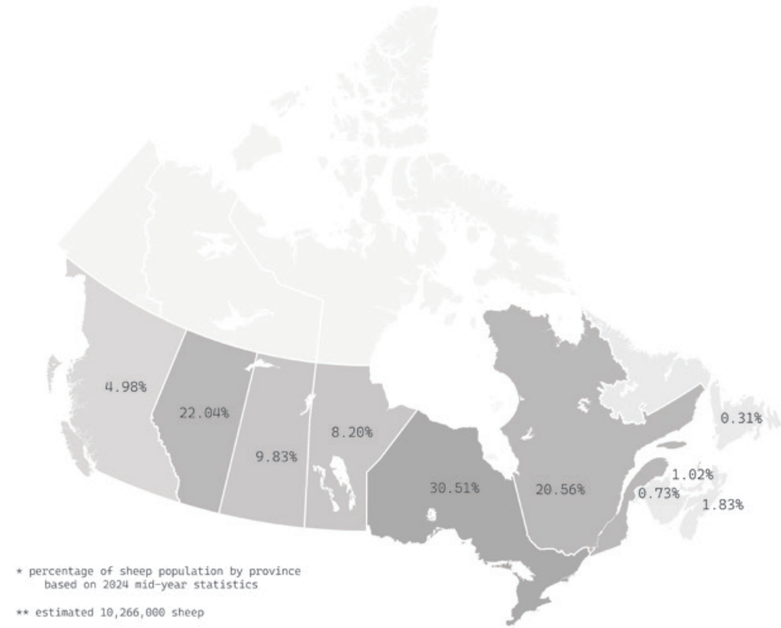
04. **Unraveling Waste, Weaving Space**

Course: ARC3018 / ARC3021 - Thesis Research + Studio
Thesis Advisor: Jeannie Kim

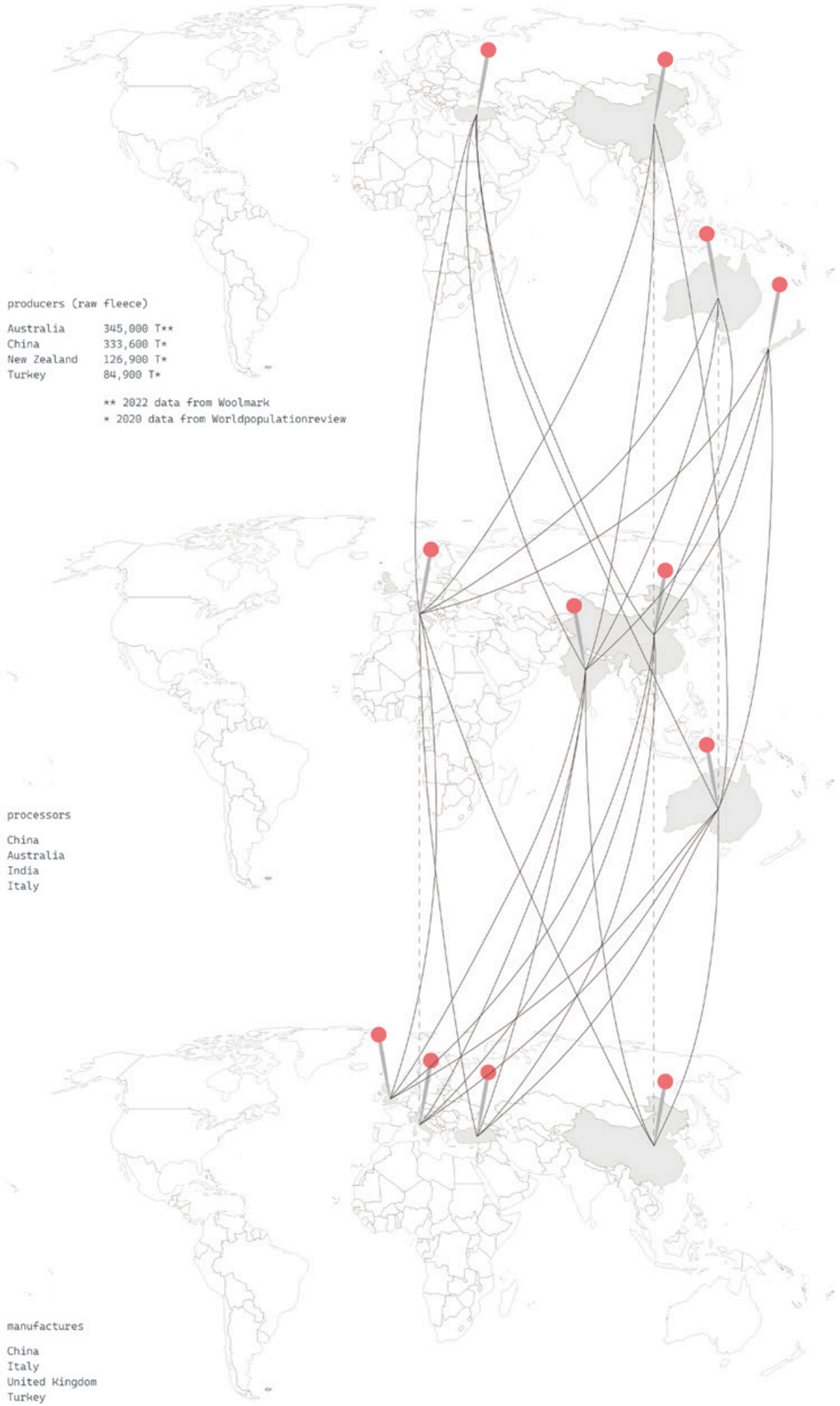
This thesis reclaims waste wool, a material once central to trade, industry, and domestic life, and reimagines it as a medium for architectural enclosure. In Canada today, most wool is discarded or burned, a byproduct of the meat and textile industries. Informed by Gottfried Semper's theory of enclosure as the origin of architecture, the project explores how the tactile, thermal, and acoustic capacities of Canadian wool might be reactivated through textile techniques such as felting, weaving, knitting, and crochet to propose a new architecture of softness.

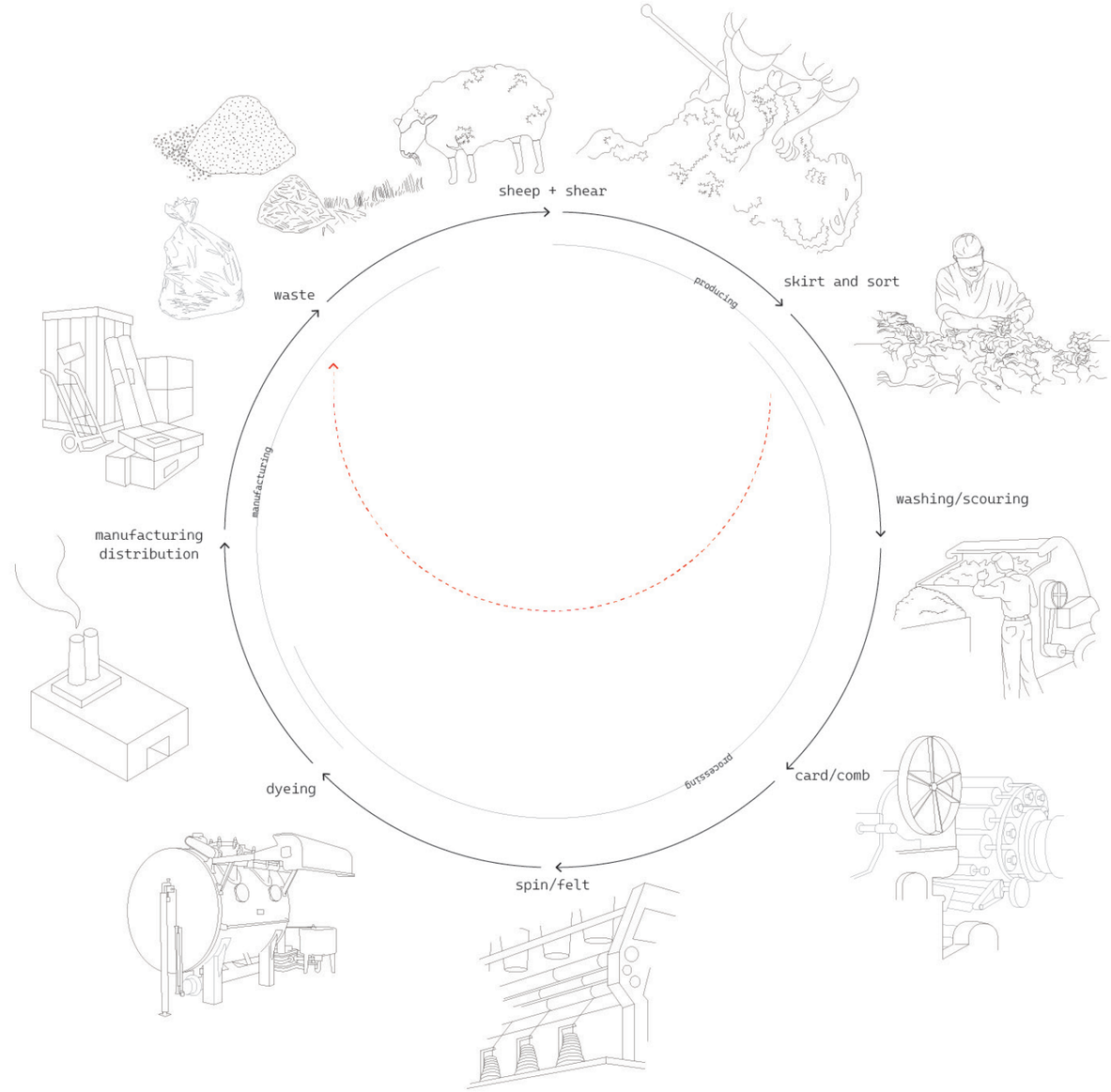
Sourced locally from a sheep farm in Guelph, Ontario, the wool used in this project becomes both the subject and material of study. By working directly with waste fleece, the research draws attention to the material's decline in value and the loss of domestic textile infrastructure that once supported it. These experiments translate foundational forms of making into architectural propositions, positioning wool not as a nostalgic craft material, but as a viable, sensorial alternative within contemporary building practice.

Following a series of hands-on studies, three prototypical sites were developed to test wool's spatial and performative potential: a bedroom, a retail ceiling, and an office meeting room. Each explores a different mode of interaction —rest, commerce, and work— revealing how a local, renewable material can move from waste to structure, from softness to system, and from the scale of the hand to the scale of architecture.



Canadian Wool Breakdown + Breeds





Wool Processing Cycle

The overlooked labour of transformation: raw fleece must be washed, dried, carded, and spun before it becomes usable, a process steeped in craft and care. This thesis investigates the often discarded portion of the cycle, between shearing/ skirting and waste, where “unusable” wool is removed, revealing broader questions around value, material standards, and potential reuse.

Global Wool Map

This map (right) traces the global network of wool production, processing, and manufacturing, illustrating the transnational journey of raw wool from sheep producing regions to industrial processing hubs and textile manufacturers. Highlighting key countries involved at each stage of the cycle, the diagram reveals the scale and complexity of wool’s commodity chain, as well as the environmental and economic distances embedded in everyday textiles.

wool locks (inverted) + resin



spun wool in resin



resin coated wool



resin bonded wool



wool locks + resin



wet felt + tie



wet felt - scallop



wet felt - insert



wet felt - strap



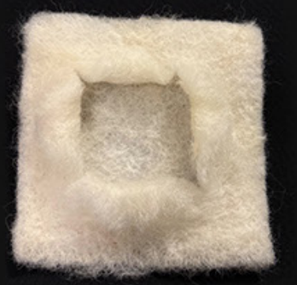
wet felt + sew



wet felt - cinch



wet felt - window



wet + needle felt form

wet felt form

needle felt roving

knit wool

wet felt - cinch

wet felt

roving + sew

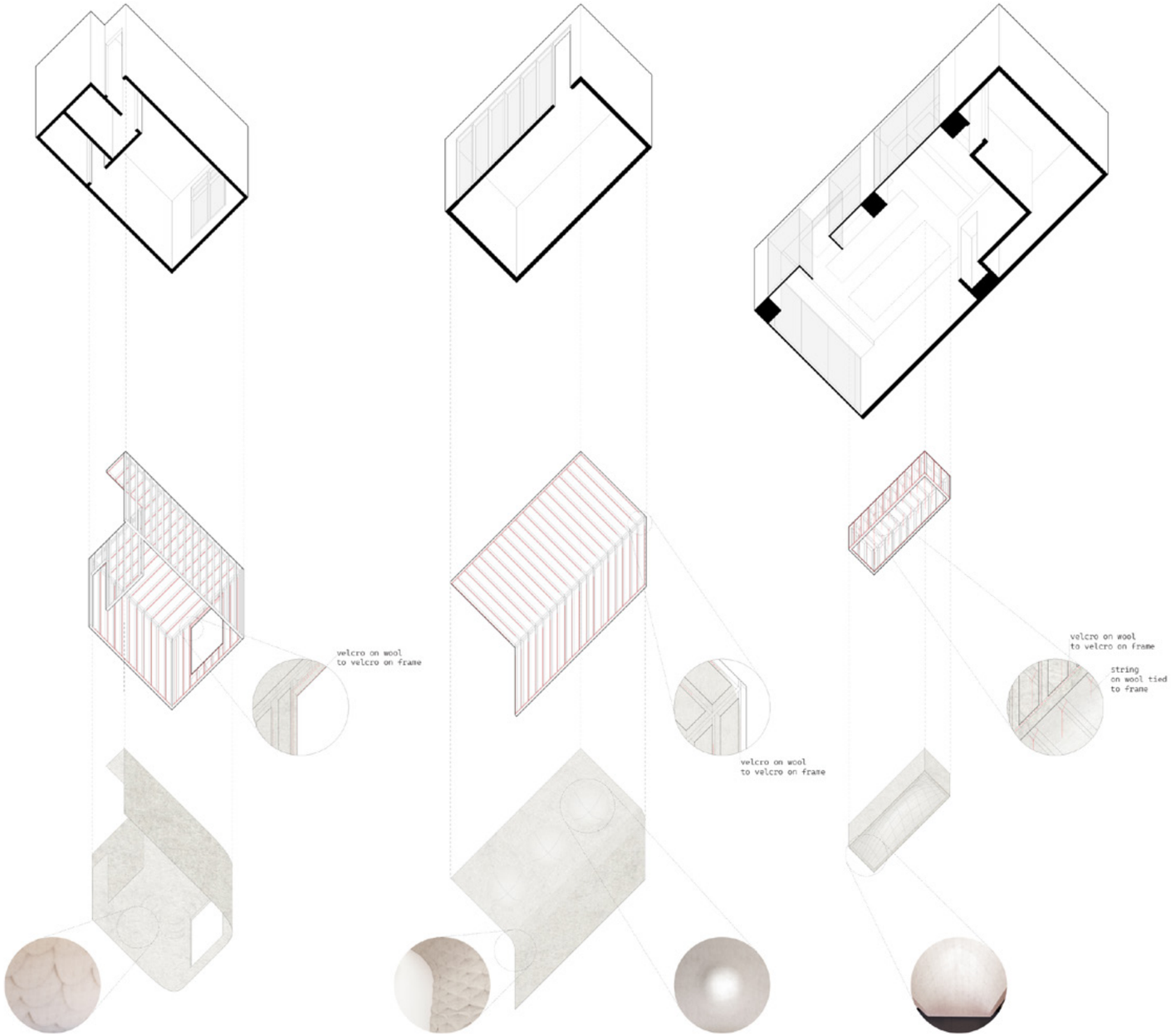
wet felt + sew

needle felt roving

needle felt

4x4 Wool Studies

Wool and composite studies pushes wool beyond the domestic and toward new material systems for structure, insulation, and enclosure. These studies offer a vocabulary of softness for architecture.



Wool-Frame System

The diagram illustrates the relationship between the wool, velcro, and cedar frame, highlighting how each material interacts within the system. The cedar frame acts as the structural backbone, with its grid providing support for the wool, which is shown in varying thicknesses to emphasize its flexibility.







Wool in Use

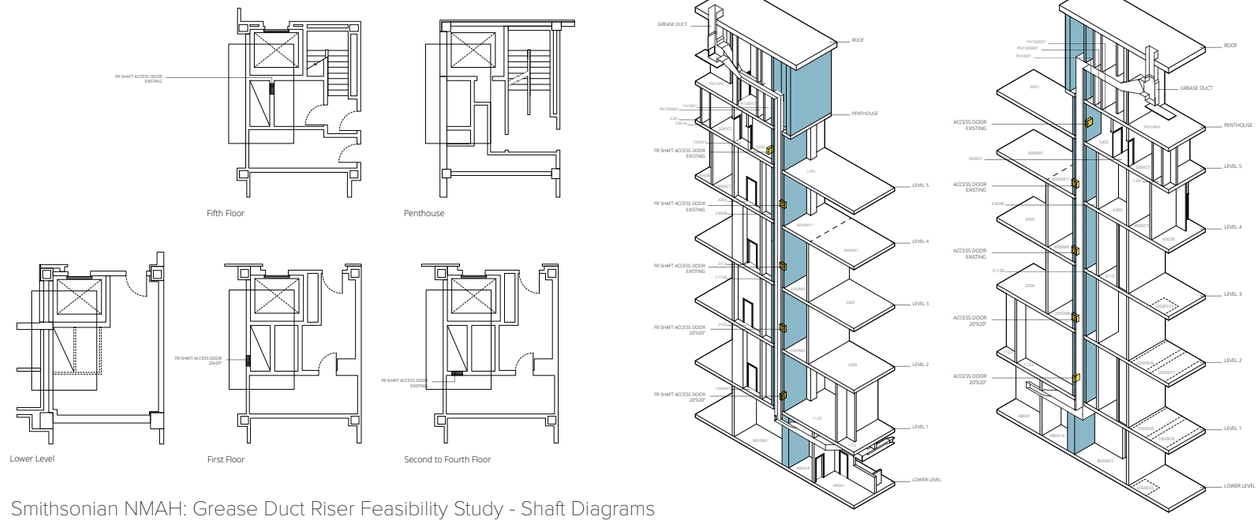
Across residential, commercial, and office settings, these prototypes reveal how waste wool can transform spaces through softness, insulation, and acoustic comfort; redefining enclosure as tactile, sensory, and restorative.

05. **Annum Architects**

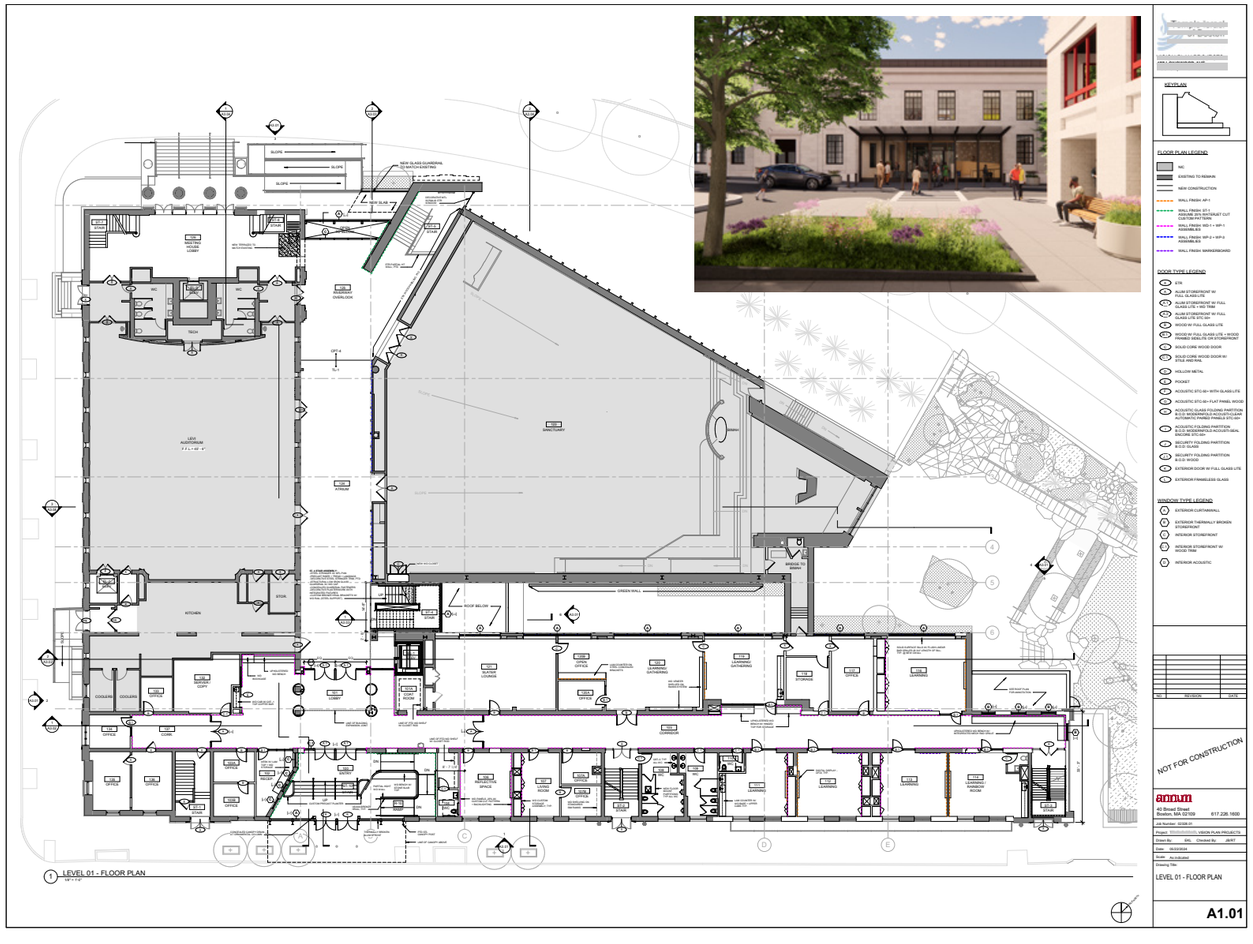
Annum Architects, formerly known as Ann Beha Architects, is a prominent architectural firm based in Boston, MA renowned for its innovative and sustainable design solutions for its work in educational institutions, cultural institutions, and historic preservation. With a commitment to integrating modern design principles with a deep respect for historical context, Annum Architects has successfully executed numerous projects that reflect a balance between contemporary aesthetics and preservation of cultural heritage.

While on my 16-month internship, I had the chance to work on part of the schematic design package and renders for a reform synagogue project in Boston, MA.

I also had the opportunity to work on pre-design, schematic design, and design development phases on the Smithsonian National Museum of American History in Washington, D.C. In addition, a few enabling projects including a grease duct riser feasibility study, emergency generator project in CD, flood strategy in CD, and AHU project in CA.



Smithsonian NMAH: Grease Duct Riser Feasibility Study - Shaft Diagrams



Reform Synagogue in Boston: SD Set and Entrance Render

thank you ☺